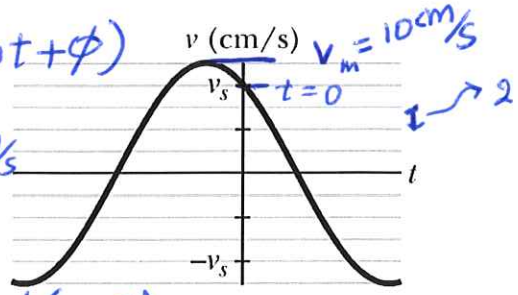
PHYS 365/369 Midterm Exam Fall 2022

1. What is the phase constant for the harmonic oscillator with the velocity function $v(t)$ given in the Figure, if the position function $x(t)$ has the form $x = x_m \cos(\omega t + \phi)$? The vertical axis scale is set by $v_s = 8.0\ cm/s$.

v_m (amplitud of the velocity)

A) +5.36 rad
B) +0.927 rad
C) +4.07 rad
D) −4.07 rad



$v = \frac{dx}{dt} = -x_m \omega \sin(\omega t + \phi)$

at $t = 0$ $\quad v = v_s = 8\ cm/s$

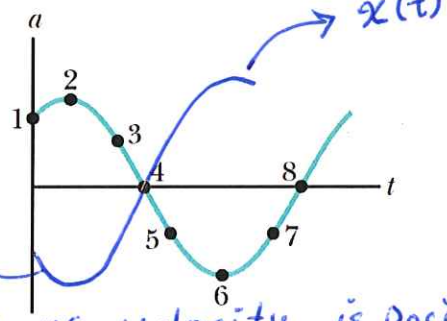
$8\ cm/s = -(x_m \omega) \sin\phi \Rightarrow \phi = \sin^{-1}(-0.8) = -0.927$ rad

OR $-0.927 + 2\pi = +5.36$ rad

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2. The acceleration $a(t)$ of a particle undergoing SHM is graphed in the Figure. Which of the. following statements is **true** about point **4**?

A) The velocity of the particle is negative
B) The particle is at $-x_m$ with velocity zero
C) The velocity of the particle is positive
D) The particle is at $+x_m$ with velocity zero



$x = x_m \cos(\omega t + \phi)$

$a = \frac{d^2x}{dt^2} = -x_m \omega^2 \cos(\omega t + \phi)$

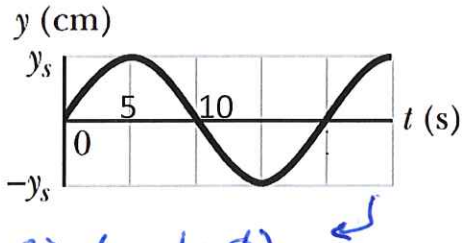
tangent to x(t) at point 4 has positive slope which means velocity is positive

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3. A sinusoidal transverse wave of wavelength 20 cm travels along a string in the positive direction of an x axis. The displacement y of the string particle at $x = 0$ is given in Figure as a function of time t. The scale of the vertical axis is set by $y_s = 10$ cm. The wave function is to be in the form of:

$y(x,t) = y_s \sin(kx - \omega t + \phi)$

A) $y(x,t) = +(10\ cm)\sin\left(\frac{\pi}{10}x - \frac{\pi}{10}t\right)$
B) $y(x,t) = -(10\ cm)\sin\left(\frac{\pi}{10}x - \frac{\pi}{10}t\right)$
C) $y(x,t) = +(10\ cm)\sin\left(\frac{\pi}{20}x - \frac{\pi}{10}t\right)$
D) $y(x,t) = -(10\ cm)\sin\left(\frac{\pi}{20}x - \frac{\pi}{10}t\right)$



The graph represents $y(0,t) = y_s \sin(-\omega t + \phi)$

at $t = 0$ $\quad y(0,0) = 0 \Rightarrow \sin(\phi) = 0$ $\begin{cases} \phi = 0 \rightarrow \text{Not acceptable because the graph is a positive sin function} \\ \phi = \pi \checkmark \end{cases}$

$y(0,t) = y_s \sin(-\omega t + \pi) = y_s \sin(\omega t)$ ✓ → positive sin function

(Next page)

3. $y(x,t) = y_s \sin(kx - \omega t + \phi) = y_s \sin(kx - \omega t + \pi)$

$= -y_s \sin(kx - \omega t) = -10 \sin\left(\frac{\pi}{10}x - \frac{\pi}{10}t\right)$

$[\sin(\theta + \pi) = -\sin(\theta)]$

$y_s = 10\,\text{cm}$, $k = \frac{2\pi}{\lambda} = \frac{2\pi}{20\,\text{cm}} = \frac{\pi}{10}$, $\omega = \frac{2\pi}{T} = \frac{\pi}{10}$

4. A longitudinal wave with an amplitude of 0.02 m moves horizontally along a Slinky with a speed of 2 m/s. Which one of the following statements concerning this wave is true?

A) Each particle in the Slinky moves a horizontal distance of 2 m each second.
B) Each particle in the Slinky moves a vertical distance of 0.04 m during each period of the wave.
C) Each particle in the Slinky moves a horizontal distance of 0.04 m during each period of the wave.
D) Each particle in the Slinky moves a horizontal distance of 0.02 m during each period of the wave.

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5. The graph shows two waves at time $t = 0$ s, one moving toward the right at 2.0 cm/s and the other moving toward the left at 2.0 cm/s. What will the amplitude be at $x = 0$ at time $t = 0.5$ s?

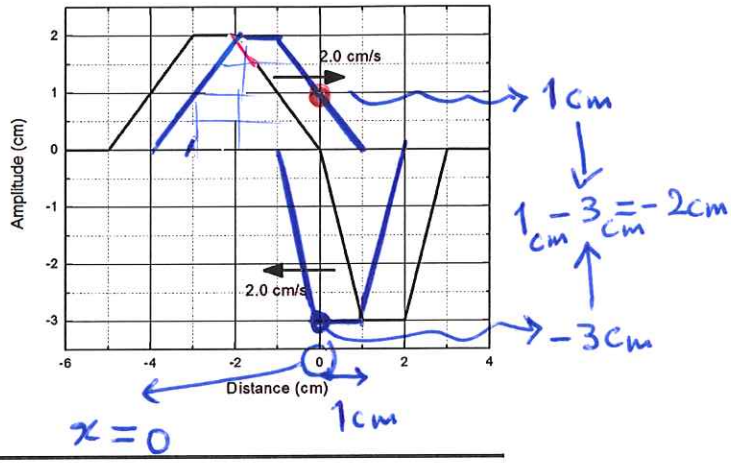
A) −3 cm
B) zero cm
C) −1 cm
D) −2 cm

$x = vt$

after $t = \frac{1}{2}$ s

$(v = 2\,\text{cm/s})$

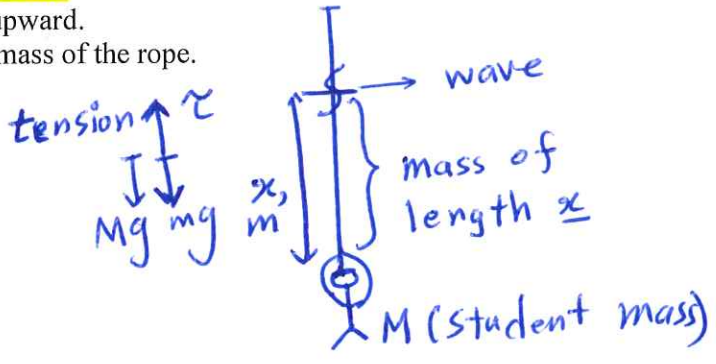
$x = 2\,\text{cm/s} \times \frac{1}{2}\,\text{s} = 1\,\text{cm}$



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6. A climbing rope is hanging from the ceiling in a gymnasium. A student grabs the end of the rope and begins moving it back and forth with a constant amplitude and frequency. A transverse wave moves up the rope. Which of the following statements describing the speed of the wave is true?

A) The speed of the wave decreases as it moves upward.
B) The speed of the wave increases as it moves upward.
C) The speed of the wave is constant as it moves upward.
D) The speed of the wave does not depend on the mass of the rope.



$\tau = mg + Mg$

$\mu = \frac{m}{x} \rightarrow m = \mu x$

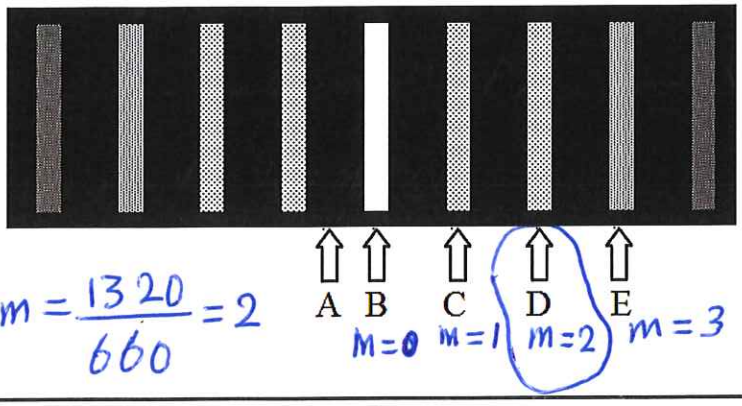
$v = \sqrt{\frac{mg + Mg}{\mu}} = \sqrt{\frac{\mu x g + Mg}{\mu}}$

$= \sqrt{\left(x + \frac{M}{\mu}\right)g}$

as x increases
speed also increases

7. A beam of monochromatic light with a wavelength of 660 nm is directed at a double slit. Consider the five fringes labeled in the drawing, including the central maximum that is labeled "**B.**" Which one of these fringes is produced when the path difference is 1320 nm?

A) A
B) C
C) D
D) E



path length difference

$\Delta L = m\lambda$

$1320_{nm} = m\,660_{nm} \Rightarrow m = \frac{1320}{660} = 2$

A: $m=0$, B: $m=1$, C: $m=2$, D: $m=3$

8. Light of wavelength 600 nm is incident upon a single slit with width 4×10^{-4} m. The figure shows the pattern observed on a screen positioned 2 m from the slits. Determine the distance s.

A) 0.002 m
B) 0.003 m
C) 0.004 m
D) 0.006 m

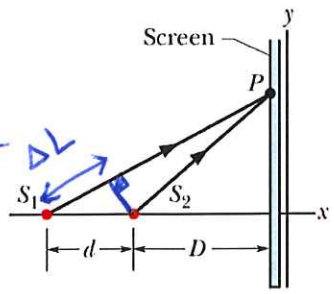


$y_m = \frac{m D \lambda}{a} = 2 \times \frac{2\,m \times 600\,nm}{4 \times 10^{-4}} = 600 \times 10^{-5}\,m = 0.006\,m$

y_m = second dark fringe ($m=2$)

9. In the Figure, two isotropic point sources S_1 and S_2 emit light in phase at wavelength λ and at the same amplitude. The sources are separated by distance $d = 6.00\,\lambda$ on an x axis. A viewing screen is at distance $D = 20.0\,\lambda$ from S_2 and parallel to the y axis. The figure shows two rays reaching point P on the screen, at height y_P. What multiple of λ gives the maximum phase difference between two rays on the screen?

A) Zero
B) 20.0
C) 6.00
D) 26



Path length difference $\leftarrow \Delta L$

ΔL becomes maximum when $\Delta L = d = 6.00\lambda$



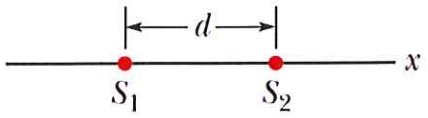
$\Delta L = d$

$m = 0, \pm 1, \pm 2$ → 8 maxima
(0: two angles; ±1: four angles; ±2: two angles)

10. In the Figure, two radiofrequency point sources S_1 and S_2, separated by distance $d = 1.0$ m, are radiating in phase with $\lambda = 0.50$ m. A detector moves in a large circular path around the two sources in a plane containing them. How many maxima does it detect?

Condition for maximum interference: $d\sin\theta = m\lambda$ (constructive)

A) 3
B) 5
C) 8
D) 10



$\sin\theta = m\frac{0.5}{1.0} = 0.5\,m$

$-1 \le \sin\theta \le +1 \Rightarrow -1 \le \frac{m}{2} \le +1 \Rightarrow -2 \le m \le 2$

Questions 11 and 12 are related to the following situation:

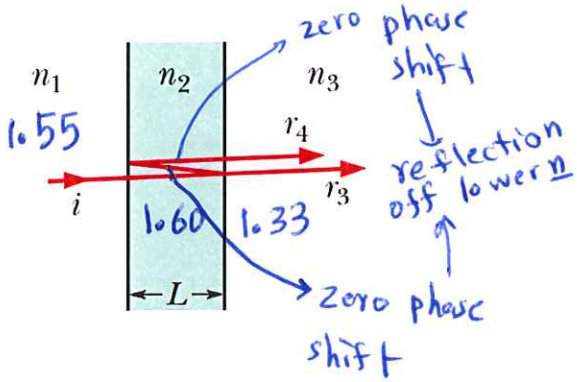
In the Figure, light is incident perpendicularly on a thin layer of material 2 that lies between (thicker) materials 1 and 3. (The rays are tilted only for clarity.) Part of the light ends up in material 3 as ray r_3 (the light does not reflect inside material 2) and r_4 (the light reflects twice inside material 2). The waves of r_3 and r_4 interfere, and here we consider the type of interference to be either maximum (max) or minimum (min). For this situation, each following questions below refers to the indexes of refraction n_1, n_2, and n_3, the type of interference, the thin-layer thickness L in nanometers, and the wavelength λ in nanometers of the light as measured in air.

Question 11 find the wavelength that is in the visible range.

Question 12 find the second least thickness.

(The visible wavelengths range from 0.4μm to 0.7μm)

Reflection off	phase shift
lower n	0
higher n	0.5 wavelength



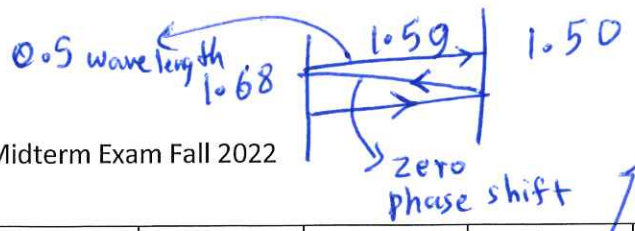
Question	n_1	n_2	n_3	Type	L	λ
11.	1.55	1.60	1.33	min	385	

A) 608 nm
B) 567 nm
C) 486 nm
D) 405 nm

Since the phase shift due to the reflection is zero: the condition for destructive interference is:

$2L = (m + \frac{1}{2})\frac{\lambda}{n_2} \rightarrow \lambda = \frac{2Ln_2}{(m+\frac{1}{2})}$, $m = 0, 1, 2, \ldots$

$\Rightarrow \lambda = \begin{cases} 1824 \text{ nm} & (m=0) \\ 608 \text{ nm} & (m=1) \rightarrow \text{invisible range} \end{cases}$



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Question	n_1	n_2	n_3	Type	L	λ
12.	1.68	1.59	1.50	Max	2nd	342

A) 376 nm
B) 269 nm
C) 161 nm
D) 54 nm

Condition for constructive interference is

$$2L = \left(m + \frac{1}{2}\right)\frac{\lambda}{n_2} \rightarrow L = \frac{2m+1}{4}\frac{\lambda}{n_2}$$

$m = 0, 1, 2, \ldots$

first least thickness

second least thickness

$m = 1 \rightarrow L = 161\,\text{nm}$

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13. An acoustical foam has a density of 20 kg/m^3. What bulk modulus should it have in order to reflect none of the sound normally incident on it? Assume the following values for air: $B_{air} = 1.42\times10^5$ Pa, $\rho_{air} = 1.2$ kg/m^3.

A) $8.5 \times 10^3\ Pa$
B) $5.8 \times 10^3\ Pa$
C) $4.7 \times 10^3\ Pa$
D) $6.5 \times 10^3\ Pa$

for normal incident

$$r = \frac{Z_1 - Z_2}{Z_1 + Z_2}$$

No reflection $\Rightarrow r = 0 \Rightarrow Z_1 = Z_2$

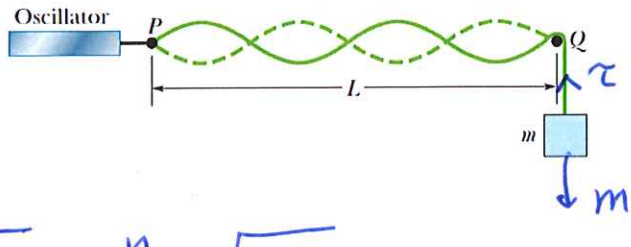
$$\Rightarrow \sqrt{\rho_{air} B_{air}} = \sqrt{\rho_f B_f} \Rightarrow B_f = \frac{\rho_{air} B_{air}}{\rho_f} = 8.5 \times 10^3\ \text{Pa}$$

(B_f: foam)

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14. In the Figure, a string, tied to a sinusoidal oscillator at P and running over a support at Q, is stretched by a block of mass m. Separation $L = 1.20$ m, linear density $\mu = 1.6$ g/m, and the oscillator frequency $f = 120$ Hz. The amplitude of the motion at P is small enough for that point to be considered a node. A node also exists at Q. What mass m allows the oscillator to set up the **fifth** harmonic on the string?

A) 0.846 Kg
B) 0.752 Kg
C) 0.597 Kg
D) 0.542 Kg

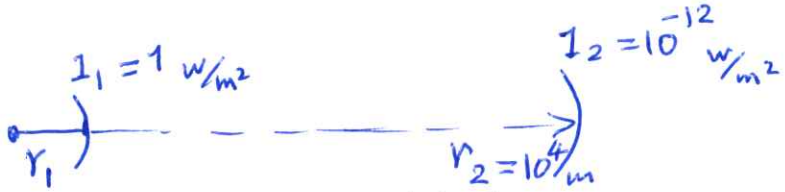


$$f = \frac{nv}{2L} = \frac{n}{2L} \times \sqrt{\frac{\tau}{\mu}} = \frac{n}{2L}\sqrt{\frac{mg}{\mu}}$$

$$\Rightarrow m = \frac{4L^2 f^2 \mu}{n^2 g} = 0.542\,\text{kg}$$

fifth harmonic

$n = 5$



15. At a distance of 10 km, a 100 Hz horn, assumed to be an isotropic point source, is barely audible. At what distance would it begin to cause pain? Table below shows some sound levels.

A) $1.0 \times 10^{-3} m$
B) $1.0 \times 10^{-2} m$
C) $1.0 \times 10^{-4} m$
D) $1.0 \times 10^{-6} m$

$I_1 = \frac{P}{4\pi r_1^2}$

$I_2 = \frac{P}{4\pi r_2^2}$

Table 17-2 Some Sound Levels (dB)

Hearing threshold	0
Rustle of leaves	10
Conversation	60
Rock concert	110
Pain threshold	120
Jet engine	130

$0 \rightarrow I_2 = 10^{-12}\ W/m^2$

$120 \rightarrow I_1 = 1\ W/m^2$

$\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2} \Rightarrow r_1 = r_2\sqrt{I_2/I_1} = 10^4\ m \times \sqrt{10^{-12}} = 10^4 \times 10^{-6}\ m = 10^{-2}\ m = 1\ cm$

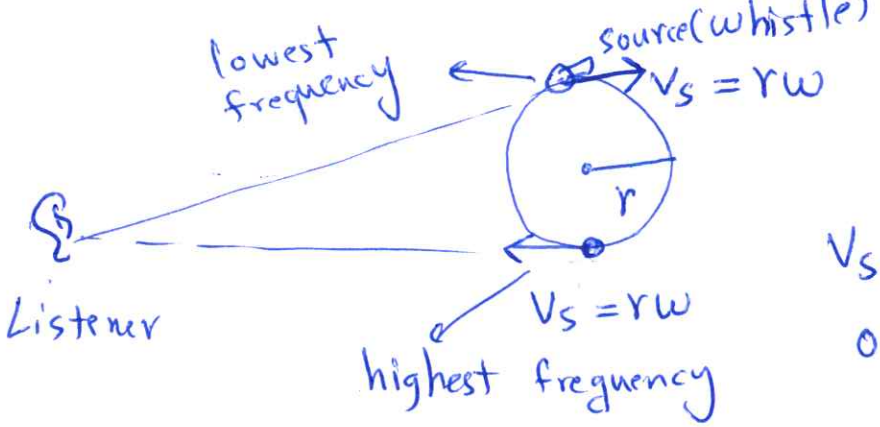
16. A whistle of frequency 550 Hz moves in a circle of radius 60 cm at an angular speed of 15.0 rad/s. What is the highest frequency heard by a listener a long distance away, at rest with respect to the center of the circle?

A) 555 Hz
B) 565 Hz
C) 525 Hz
D) 536 Hz

$f_L = \frac{V \pm V_L}{V \mp V_S} f_L$ (upper sign for motion toward)

f_L: listener

V_L: 0 speed of listener is zero



$V_S = rw = 0.6\ m \times 15\ rad/s = 9\ m/s$

$f_L = \frac{V}{V - V_S} f_L = \frac{343}{343 - 9} \times 550\ Hz = 565\ Hz$